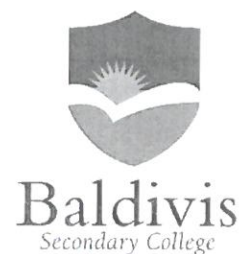


Year 10
Pathway 1 and 2



Mini Test 1 (version 2)
Indices and Algebraic fractions

Name: ANSWERS

Score: /44
Total Time: 45 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed.

Simplifying and Expanding			
1 Simplify $4a - 2b + 2a + 8b$	[1]	2 Expand $5(2t + 1)$	[1]
$6a + 6b$		$10t + 5$	
3 Expand $-3x(x - 4)$	[1]	4 Expand $(x + 7)(x + 1)$	[1]
$-3x^2 + 12x$		$x^2 + 8x + 7$	
5 Expand $(x + 1)(x - 4)$	[1]	6 Expand $(a - 5)^2$	[2]
$x^2 - 3x - 4$		$(a - 5)(a - 5)$ $= a^2 - 10a + 25$	
7 Expand $(3p - 2)(4p + 9)$	[2]	8 Expand $(2x + 3)^2 - 3(x + 1)$	[2] 3
$12p^2 + 27p - 8p - 18$ $= 12p^2 + 19p - 18$		$(2x + 3)(2x + 3) - 3x + 3$ $= 4x^2 + 12x + 9 - 3x + 3$ $= 4x^2 + 9x + 12$	

Index Laws

9 Evaluate 5^5 [1] 3125	10 Evaluate 6^0 [1] 1
11 Simplify $4^4 \times 7$ leaving your answer in index form. [1] $4^4 \times 7$	12 Simplify $x^2 \div x^9$ Leaving your answer with positive indices [1] $x^{-7} = \frac{1}{x^7}$
13 Simplify $12x^6 \div 3x^3$ [1] $4x^3$	14 Simplify $(y^{-5})^7$ [1] y^{-35}
15 Simplify $(3a^4b)^5$ [2] $3^5 a^{4 \times 5} b^{1 \times 5}$ $= 243a^{20}b^5$	16 Simplify and leave your answer in index form [2] $\frac{2^7 \times 3^4 \times 5^9}{5^2 \times 2^3}$ $= 2^4 \times 3^4 \times 5^7$
17 Simplify and leave your answer in index form [3] $\frac{(3^3 \times 2^5)^3}{2^{11} \times 3}$ $\frac{3^{3 \times 3} \times 2^{5 \times 3}}{2^{11} \times 3} = \frac{3^9 \times 2^{15}}{2^{11} \times 3}$ $= 3^8 \times 2^4$	18 Simplify and leave your answer in index form [2] $7p^2q^5 \div 42p^2q^4$ $7p^2q^4(2+6)$

Algebraic Fractions and Factorising

19 Factorise $16x + 24wx^3$

[2]

$$8x(2 + 3wx^2)$$

21 Simplify and leave your answer in index form

[3]

$$\left(\frac{5}{8^2}\right)^4 \times \left(\frac{4^2}{5}\right)^5$$

$$\frac{5^4}{8^8} \times \frac{4^{10}}{5^5} = \frac{5^{-1} \times 4^{10}}{8^8 \times 5^5}$$

$$= \frac{4^{10}}{8^8 \times 5}$$

20 Simplify and leave your answer in index form

[2]

$$\frac{t+5}{12} + \frac{4t}{6}$$

$$\frac{t+5}{12} + \frac{8t}{12}$$

$$= \frac{9t+5}{12}$$

21 Simplify and leave your answer in index form

[1]

$$\frac{x}{2} + \frac{4x}{2}$$

$$\frac{5x}{2}$$

22 Simplify

[3]

$$\frac{4x+5}{5} + \frac{x-8}{5}$$

$$\frac{5x-3}{5}$$

23 Simplify

[1]

$$\frac{v}{3} \times \frac{4w}{5}$$

$$\frac{4wv}{15}$$

24 Simplify

[3]

$$\frac{3x}{8} + \frac{x}{3}$$

$$\frac{9x}{24} + \frac{8x}{24}$$

$$= \frac{17x}{24}$$

25 The product of x and y is added to the reciprocal of the sum of x and y .

(a) Write down the expression that describes this information [1]

$$xy + \frac{1}{x+y}$$

(b) Write the expression in part (a) with a common denominator. [3]

$$\frac{xy}{1} \times \frac{(x+y)}{(x+y)} + \frac{1}{(x+y)}$$
$$= \frac{x^2y + xy^2}{(x+y)} + \frac{1}{x+y}$$

$$= \frac{x^2y + xy^2 + 1}{x+y}$$

End of Test